Boundary Charges, One-Loop Renormalisation, and Obstructions to Quantum Defect Fusion for the Corrigan–Zambon Boussinesq Defect

Project *quantum_ boussinesq_ solitons* (Route E, next step #3) — PI: Gustav Delius

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Abstract

We carry out ranked next step #3 of the revised Phase 2 master report (`phase2/report.pdf`): deriving the localised classical and one-loop boundary charges and renormalisation structure of the Corrigan–Zambon (CZ) type-I integrable defect [1] for the good Boussinesq equation before making any physical claim of quantum defect fusion. Every nontrivial claim is labelled **(a)** (literature), **(b)** (literature heuristic), **(c)** (derived and verified in this project), or **(d)** (guess/open physical identification).

Our main results **(c)** are:

1. **Classical localised defect charges and exact conservation laws (`phase2/scripts/defq_classical_charges.py`).** Starting from the total action in fixed project conventions ($u_{\mathrm{CZ}} = -3\varphi$, $\mathcal{L}_{\mathrm{CZ}} = 9\mathcal{L}$), we derive the four nonlinear sewing conditions and the exact localised defect energy E_D, modified momentum P_D, φ-shift charge Π_D, and topological mass jump $M_D = \varphi_R(0^+) - \varphi_L(0^-)$. We prove symbolically that $\frac{\mathrm{d}}{\mathrm{d}t}(E_{\mathrm{bulk}} + E_D) = 0$, $\frac{\mathrm{d}}{\mathrm{d}t}(P_{\mathrm{bulk}} + P_D) = 0$, $\frac{\mathrm{d}}{\mathrm{d}t}(\Pi_{\mathrm{bulk}} + \Pi_D) = 0$, and $\frac{\mathrm{d}}{\mathrm{d}t}(M_{\mathrm{bulk}} + M_D) = 0$ hold off-shell on the sewing manifold, that the vacuum jumps across any KP triple edge $S_i = \{q_i, q_{i+1}\}$ factorise modulo $q_i^2 + q_i q_{i+1} + q_{i+1}^2 = \frac{1}{4}$ into $\Delta(E_D, P_D, \Pi_D, M_D)_{q_i \to q_{i+1}} = \rho_i(E, P, \Pi, M)_{S_i}$ with $\rho_i = \eta\,\mathrm{sgn}(q_i - q_{i+1})$, and we verify time-independent total charges to 10^{-50} on exact transmitting and absorbing soliton–defect solutions.

2. **Corrected sewing determinant, admissible growing-mode branch switch, and exact spectral trace of $P_{D,2}$ (`phase2/scripts/defq_oneloop_renorm.py`).** Incorporating the independent audit (`audit_frames_defects.pdf`), we verify symbolically that the reduced 4×4 linearised sewing determinant carries the prefactor $-\frac{2}{3}$ (sixteen times the Phase 2g prefactor $-\frac{1}{24}$) and that its continuum-threshold values factorise as $P_-(1; s) = \frac{1}{2}(s+1)(s-\sqrt{3})$ and $P_+(1; s) = \frac{1}{2}(s-1)(s-\sqrt{3})$ ($s = \eta\sigma_{\mathrm{vac}}$). This proves an exact admissible-branch switch ($S = \kappa_1 + \kappa_2 > 1$) at $s = +\sqrt{3}$, where $S_{-,+}(s)$ exits the physical sheet and $S_{+,-}(s)$ enters from P_+, before coalescing with $S_{+,+}(s)$ at $s = 2$. On the reflectionless continuum phonon basis, we prove the **exact pointwise identity**

$$\langle P_{D,2}\rangle_q + \langle P_{D,2}\rangle_{-q} \equiv \frac{q}{4\pi}\Big(\theta_D'(q; s, \eta) - \theta_D'(-q; s, \eta)\Big),$$

resolving the open question in `audit_charges_action.pdf` by showing directly from the quadratic boundary operator $P_{D,2}$ that $\int_{-\infty}^{\infty} \mathrm{d}q\, \langle P_{D,2}\rangle_q = \Delta P_D(q_D, \eta)$. Similarly, Euler's theorem on the quadratic sewing Lagrangian $D_2 = a_t F_1 + b_t G_1 - E_{D,2}$ proves that $\langle E_{D,2} + J_{\mathrm{bulk},2}\rangle_q = \langle a_t F_1 + b_t G_1\rangle_q$ and $-\big(\langle a_t F_1 + b_t G_1\rangle_q + \langle a_t F_1 + b_t G_1\rangle_{-q}\big) = \frac{\omega(q)}{4\pi}\big(\theta_D'(q) + \theta_D'(-q)\big)$.

3. **Boundary UV renormalisation and three sharp obstructions to physical quantum defect fusion (`phase2/scripts/defq_absorption_fusion.py`).** With equal-mode-count cutoff Λ in the asymptotic phase gauge $\theta_D(\pm\infty) = 0$, the Casimir energy $E_D^{\mathrm{mode}}(\Lambda) = E_D^{\mathrm{dens}}(\Lambda) - B_D(\Lambda)$ diverges linearly as $\frac{S_D}{\pi}(\sqrt{1+\Lambda^2} - 1) + \Delta E_D^{\mathrm{ren}} + O(\Lambda^{-1})$ with $S_D = -\frac{\eta\sqrt{3}\sigma_{\mathrm{vac}}}{2} = 3\eta q_D$. Because the bulk normal-ordering counterterm

$\propto \int_{x\neq 0} \varphi_x \, \mathrm{d}x$ vanishes on constant defect vacua, cancelling the defect UV divergence requires either a distributional boundary tadpole counterterm $\propto (\varphi_R(0^+) - \varphi_L(0^-))$ or a boundary renormalisation of the defect parameters (s, ρ_0, τ_0, h_0). Across KP triples we establish three obstructions to physical $O(\hbar^0)$ triangle fusion:

(i) *Stability obstruction:* Every nondegenerate KP triple has growing-mode counts $(0, 1, 2)$ generically (and $(0, 1, 1)$ at the branch-switch locus $s_{\text{eff}} = \sqrt{3}$), so **zero** single-soliton absorption/emission edges connect two linearly stable defect vacua.

(ii) *Canonical-charge holonomy no-go theorem:* While the mode-sum charges $(\Delta E_1^{\text{ren}}, \Delta P_{\text{mode}})$ have zero directed sum around every KP triangle, the co-moving canonical collective-coordinate charges $(\Delta E_{\text{can}}, \Delta P_{\text{can}})$ have a **non-zero triangle holonomy** $\sum_{i=1}^{3} \rho_i(\Delta E_{\text{can}}(S_i), \Delta P_{\text{can}}(S_i)) \neq (0, 0)$. Hence *no* single-valued assignment of renormalised defect vacuum charges $(E_D^{\text{quant}}(q_i), P_D^{\text{quant}}(q_i))$—regardless of boundary counterterms—can balance $(\Delta E_{\text{can}}, \Delta P_{\text{can}})$ at unshifted classical absorption rapidities across all three edges of a KP triangle.

(iii) *Absorbed-soliton momentum-renormalisation obstruction:* Holding both soliton physical momenta $(P_3^{\text{phys}}, P_{S_i}^{\text{phys}})$ fixed in $S(3, S_i)$ adds the second Taylor term $\Delta X_{S_i}(3, S_i)\Delta P(S_i)$, whose directed sum around a KP triangle equals $\sum_{i=1}^{3} \left(\frac{\Delta P(S_{i-1})}{k_{S_{i-1}}} - \frac{\Delta P(S_i)}{k_{S_i}}\right) \ln|z_t(3; q_i, \eta)| \neq 0$ for both ΔP_{mode} and ΔP_{can}.

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1 Conventions, CZ dictionary, and scope

We work throughout in the fixed project conventions (`AGENTS.md`, `README.md`) for the **good** Boussinesq equation **(a)**:

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \varphi_x = 2(\ln\tau)_{xx}, \tag{1}$$

with bulk Lagrangian density on the left ($x < 0$, φ_L) and right ($x > 0$, φ_R) half-lines and canonical symplectic form

$$\mathcal{L}(\varphi) = \frac{1}{2}\varphi_t^2 - \frac{1}{2}\varphi_x^2 + \varphi_x^3 - \frac{1}{2}\varphi_{xx}^2, \qquad \Omega_{\text{bulk}} = \int_{-\infty}^{0} \mathrm{d}\pi_L \wedge \mathrm{d}\varphi_L + \int_{0}^{\infty} \mathrm{d}\pi_R \wedge \mathrm{d}\varphi_R, \quad \pi_{L,R} = \varphi_{L,R,t}. \tag{2}$$

Planck's constant $\hbar$ refers to this normalisation.

A one-soliton with amplitude parameter $\alpha \in (0, \pi/2)$ and propagation direction $\epsilon \in \{+1, -1\}$ has **(a)**/**(c)**:

$$k = \sin\alpha, \quad v = \epsilon\cos\alpha, \quad \ell = -\epsilon\cos 3\alpha, \quad M = 2k,$$
$$\Pi = -Mv, \quad P = \frac{2}{3}\epsilon k^3\cos\alpha, \quad E = \frac{2}{15}k^3(5-4k^2), \tag{3}$$

and KP roots $p_a > p_b$ satisfying $p_a - p_b = k$, $p_a + p_b = v/\sqrt{3}$, and $4p^3 - p = -\frac{\epsilon}{3\sqrt{3}}\cos 3\alpha = \frac{\ell}{3\sqrt{3}}$ **(c)** (`defect_semiclassics.pdf`).

Corrigan–Zambon dictionary. Corrigan and Zambon (CZ) [1] (with $c = \lambda = \kappa = 1$) write $u_{\rm CZ}$ for $x < 0$ and $v_{\rm CZ}$ for $x > 0$ with bulk Lagrangian $\mathcal{L}_{\rm CZ} = \frac{1}{2}u_{{\rm CZ},t}^2 - \frac{1}{2}u_{{\rm CZ},x}^2 - \frac{1}{3}u_{{\rm CZ},x}^3 - \frac{1}{2}u_{{\rm CZ},xx}^2$. The explicit dictionary to project variables is **(c)** (`phase2/scripts/defq_classical_charges.py`):

$$u_{\rm CZ} = -3\varphi_L, \qquad v_{\rm CZ} = -3\varphi_R, \qquad \mathcal{L}_{\rm CZ} = 9\,\mathcal{L}, \qquad D_{\rm proj} = \frac{1}{9}D_{\rm CZ}, \qquad (E_{\rm CZ}, P_{\rm CZ}) = 9\,(E, P). \tag{4}$$

We do not rely on any results from Phase 2i (`toda_wh_and_kp_synthesis.pdf`), which is under separate audit.

2 Classical localised defect charges and exact conservation laws

2.1 Sewing conditions and localised charges in project conventions

In CZ variables [1] **(a)**, the type-I defect at $x = 0$ has orientation $\eta \in \{+1, -1\}$, $\alpha_{\rm CZ} = \eta/\sqrt{3}$, $\beta_{\rm CZ} = 2\alpha_{\rm CZ} = 2\eta/\sqrt{3}$, $\mu = \frac{2}{3\beta_{\rm CZ}} = \eta/\sqrt{3}$, and bare parameter $s = \sigma_0 \in \mathbb{R}$ (with KP vertex $q_D = -\frac{s}{2\sqrt{3}}$). In terms of the potential jump at $x = 0$,

$$w_\varphi := \varphi_L(0^-, t) - \varphi_R(0^+, t) = -M_D, \qquad w_{\rm CZ} = u_{\rm CZ}(0^-, t) - v_{\rm CZ}(0^+, t) = -3w_\varphi = 3M_D, \tag{5}$$

the dynamical defect field σ and auxiliary polynomials $\rho(\sigma)$, $\tau(\sigma)$, and $h(\sigma)$ are **(a)**/**(c)** (`phase2/scripts/defq_classical_charges.py`):

$$\sigma(w_\varphi) = s + \mu w_{\rm CZ} = s - \sqrt{3}\,\eta\,w_\varphi, \qquad \rho_0 = -\frac{\sqrt{3}\,\eta}{2}\,s\left(1 - \frac{s^2}{3}\right) = \frac{\eta}{\sqrt{3}}\,\ell_D, \tag{6}$$

$$\rho(\sigma) = \rho_0 + \frac{\sqrt{3}\,\eta}{2}\left(\sigma - \frac{\sigma^3}{3}\right) = \frac{\sqrt{3}\,\eta}{6}\,(\sigma - s)\big(3 - (\sigma^2 + s\sigma + s^2)\big), \tag{7}$$

$$\tau(\sigma) = -\int_s^\sigma \frac{\beta_{\rm CZ}}{2\mu}\,x\,\rho(x)\,{\rm d}x = -\int_s^\sigma x\,\rho(x)\,{\rm d}x, \tag{8}$$

$$h(\sigma) = -\int_s^\sigma \frac{\beta_{\rm CZ} - \alpha_{\rm CZ}}{\mu}\,\rho(x)\,{\rm d}x = -\int_s^\sigma \rho(x)\,{\rm d}x. \tag{9}$$

In the shift-invariant gauge $f = g = -\frac{\sigma^2}{4\mu} = -\frac{\sqrt{3}\eta}{4}\sigma^2$ (which satisfies $f_{v_{\rm CZ}} - g_{u_{\rm CZ}} = \sigma$ and preserves $\varphi \mapsto \varphi + c$ **(c)**, `defect_semiclassics.pdf`), the project-normalised defect Lagrangian at $x = 0$ is

$$D_{\rm proj} = \varphi_{L,t}F_{\rm proj} + \varphi_{R,t}G_{\rm proj} - E_D, \tag{10}$$

where **(c)** (`phase2/scripts/defq_classical_charges.py`):

$$F_{\rm proj} = \frac{\eta}{\sqrt{3}}\big(\varphi_{L,x} + 2\varphi_{R,x}\big) + \frac{\sqrt{3}\,\eta}{12}\,\sigma^2, \qquad G_{\rm proj} = \frac{\eta}{\sqrt{3}}\big(2\varphi_{L,x} + \varphi_{R,x}\big) + \frac{\sqrt{3}\,\eta}{12}\,\sigma^2, \tag{11}$$

$$E_D = -\frac{H_{\rm CZ}}{9} = -\frac{\eta}{\sqrt{3}}\,\sigma\big(\varphi_{L,x}^2 + \varphi_{L,x}\varphi_{R,x} + \varphi_{R,x}^2\big) + \frac{1}{3}\,\rho(\sigma)\big(\varphi_{L,x} + \varphi_{R,x}\big) - \frac{1}{9}\,\tau(\sigma). \tag{12}$$

Proposition 2.1 (Classical localised defect charges and off-shell conservation, **(c)**, `phase2/scripts/defq_classical_charges.py`). *Varying* $S = \int_{x<0} \mathcal{L}(\varphi_L)\,\mathrm{d}x\,\mathrm{d}t + \int_{x>0} \mathcal{L}(\varphi_R)\,\mathrm{d}x\,\mathrm{d}t + \int D_{\mathrm{proj}}\,\mathrm{d}t$ *yields the four sewing conditions at* $x = 0$*:*

$$\varphi_{L,xx} = \frac{\partial D_{\mathrm{proj}}}{\partial \varphi_{L,x}}, \qquad \varphi_{L,xxx} = \varphi_{L,x} - 3\varphi_{L,x}^2 - \frac{\partial D_{\mathrm{proj}}}{\partial \varphi_L} + \frac{\mathrm{d}F_{\mathrm{proj}}}{\mathrm{d}t}, \tag{13}$$

$$\varphi_{R,xx} = -\frac{\partial D_{\mathrm{proj}}}{\partial \varphi_{R,x}}, \qquad \varphi_{R,xxx} = \varphi_{R,x} - 3\varphi_{R,x}^2 + \frac{\partial D_{\mathrm{proj}}}{\partial \varphi_R} - \frac{\mathrm{d}G_{\mathrm{proj}}}{\mathrm{d}t}. \tag{14}$$

For any smooth solution of the bulk equation (1) *on* $x \neq 0$ *satisfying* (13)–(14) *at* $x = 0$*, the four total charges*

$$E_{\mathrm{tot}} = E_{\mathrm{bulk}} + E_D, \quad P_{\mathrm{tot}} = P_{\mathrm{bulk}} + P_D, \quad \Pi_{\mathrm{tot}} = \Pi_{\mathrm{bulk}} + \Pi_D, \quad M_{\mathrm{tot}} = M_{\mathrm{bulk}} + M_D, \tag{15}$$

are conserved identically ($\frac{\mathrm{d}}{\mathrm{d}t}(E_{\mathrm{tot}}, P_{\mathrm{tot}}, \Pi_{\mathrm{tot}}, M_{\mathrm{tot}}) = (0,0,0,0)$)*, where* E_D *is given by* (12) *and the localised defect momentum* P_D*, shift charge* Π_D*, and mass jump* M_D *in project normalisation are:*

$$P_D = \frac{P_{D,\mathrm{CZ}}}{9} = -\frac{\eta}{2\sqrt{3}}\big(\varphi_{L,x}^2 + 4\varphi_{L,x}\varphi_{R,x} + \varphi_{R,x}^2\big) + \frac{1}{9}\,h(\sigma), \tag{16}$$

$$\Pi_D = F_{\mathrm{proj}} + G_{\mathrm{proj}} = \sqrt{3}\,\eta\big(\varphi_{L,x} + \varphi_{R,x}\big) + \frac{\sqrt{3}\,\eta}{6}\,\sigma^2, \tag{17}$$

$$M_D = \varphi_R(0^+,t) - \varphi_L(0^-,t) = -w_\varphi = \frac{\sigma - s}{\sqrt{3}\,\eta}. \tag{18}$$

Remark 2.2 (Avoiding double-subtraction in P_D, **(c)**). In CZ [1, (3.11)], the modified momentum is introduced in the intermediate form $P_{\mathrm{bulk,CZ}} - u_{\mathrm{CZ},x}F_{\mathrm{CZ}} - v_{\mathrm{CZ},x}G_{\mathrm{CZ}} + \tilde{P}_D$, and in their final formula [1, (3.49)] the derivative terms $-u_{\mathrm{CZ},x}F_{\mathrm{CZ}} - v_{\mathrm{CZ},x}G_{\mathrm{CZ}}$ have already been combined with $\tilde{P}_D$ to produce $P_{D,\mathrm{CZ}} = -\frac{\alpha_{\mathrm{CZ}}}{2}(u_{\mathrm{CZ},x}^2 + v_{\mathrm{CZ},x}^2) - \beta_{\mathrm{CZ}}u_{\mathrm{CZ},x}v_{\mathrm{CZ},x} + h(\sigma)$, which is free of time derivatives. Subtracting $u_{\mathrm{CZ},x}F_{\mathrm{CZ}} + v_{\mathrm{CZ},x}G_{\mathrm{CZ}}$ a second time leaves a non-zero time-derivative remainder (recorded in Section 5, Failure 1).

2.2 Exact conservation through soliton absorption $D(q_i) + S_i \to D(q_{i+1})$

On a static defect vacuum ($\varphi_{L,x} = \varphi_{R,x} = 0$), the sewing condition (13) requires $\rho(\sigma) = 0$, whose three roots $\sigma_i = -2\sqrt{3}\,q_i$ correspond to the three roots $\{q_1, q_2, q_3\}$ of the KP cubic $q_i^2 + q_iq_j + q_j^2 = \frac{1}{4}$ containing $q_D = -\frac{s}{2\sqrt{3}}$ **(c)** (`defect_semiclassics.pdf`). Evaluating (12) and (16)–(18) on $\varphi_{L,x} = \varphi_{R,x} = 0$ gives the vacuum charges of vertex q_i **(c)** (`phase2/scripts/defq_classical_charges.py`):

$$E_D^{\mathrm{vac}}(q_i) = -\frac{1}{9}\tau(\sigma_i),\ P_D^{\mathrm{vac}}(q_i) = \frac{1}{9}h(\sigma_i),\ \Pi_D^{\mathrm{vac}}(q_i) = \frac{\sqrt{3}\,\eta}{6}\sigma_i^2 = 2\sqrt{3}\,\eta\,q_i^2,\ M_D^{\mathrm{vac}}(q_i) = 2\eta(q_D - q_i). \tag{19}$$

Proposition 2.3 (Symbolic and 50-digit time-dependent conservation in absorption, **(c)**, `phase2/scripts/defq_classical_charges.py`). *Let* $\{q_1, q_2, q_3\}$ *be a KP triple* ($q_i^2 + q_iq_{i+1} + q_{i+1}^2 = \frac{1}{4}$)*,* $S_i = \{q_i, q_{i+1}\}$ *the soliton with KP roots* $p_a = \max(q_i, q_{i+1})$ *and* $p_b = \min(q_i, q_{i+1})$*, and* $\rho_i := \eta\,\mathrm{sgn}(q_i - q_{i+1}) \in \{+1, -1\}$*.*

1. *Modulo the KP triple polynomial* $q_i^2 + q_iq_{i+1} + q_{i+1}^2 - \frac{1}{4} = 0$*, the vacuum jumps of all four defect charges* (19) *equal* ρ_i *times the classical soliton charges* (3) *identically:*

$$\boxed{\big(E_D^{\mathrm{vac}}, P_D^{\mathrm{vac}}, \Pi_D^{\mathrm{vac}}, M_D^{\mathrm{vac}}\big)\big|_{q_i}^{q_{i+1}} = \rho_i\,\big(E(S_i),\,P(S_i),\,\Pi(S_i),\,M(S_i)\big).} \tag{20}$$

2. *On the exact time-dependent CZ one-soliton solutions—both in regular transmission* ($z > 0$) *and in exact absorption* $D(q_1) + S_1 \to D(q_2)$ ($z = 0$ *on* $x > 0$)*—direct spatial quadrature of* $(E_{\mathrm{bulk}}(t), P_{\mathrm{bulk}}(t), \Pi_{\mathrm{bulk}}(t), M_{\mathrm{bulk}}(t))$ *plus the localised defect charges* $(E_D(t), P_D(t), \Pi_D(t), M_D(t))$ *at* $x = 0$ *is independent of* $t \in [-8, +8]$ *to* 9.35×10^{-51} *(transmission) and* 5.35×10^{-51} *(absorption).*

3 One-loop defect charges, spectral branches, and boundary renormalisation

3.1 Corrected sewing determinant and the admissible branch switch at $s_{\rm eff} = \sqrt{3}$

Linearising the sewing conditions (13)–(14) around a static defect vacuum $\sigma_{\rm vac}$ with effective parameter $s_{\rm eff} := \eta\sigma_{\rm vac} = -2\sqrt{3}\,\eta\, q_{\rm vac}$ and seeking localised growing modes of frequency $\omega = {\rm i}g$ ($g > 0$) and spatial decay exponents $\kappa_{1,2} > 0$ ($\kappa_1^2 + \kappa_2^2 = 1$, $\kappa_1\kappa_2 = g$, $S := \kappa_1 + \kappa_2 = \sqrt{1+2g} > 1$, $d := \kappa_1 - \kappa_2 = \sqrt{2 - S^2}$) yields a 4×4 sewing matrix $M_{4\times4}(g; \sigma_{\rm vac}, \eta)$. Following the independent audit `audit_frames_defects.pdf`, we verify its exact off-shell polynomial factorisation and determine the complete admissible branch structure **(c)** (`phase2/scripts/defq_oneloop_renorm.py`).

Proposition 3.1 (Corrected sewing determinant and admissible branch switch, **(c)**, `phase2/scripts/defq_or`

Off-shell for all $(S, s_{\rm eff})$, the reduced 4×4 linearised sewing determinant is

$$\frac{\det M_{4\times4}(g; \sigma_{\rm vac}, \eta)}{(\kappa_1 - \kappa_2)^2} = -\frac{2}{3}\, S^2(S^2 - 1)\, P_-(S; s_{\rm eff})\, P_+(S; s_{\rm eff}), \tag{21}$$

where $P_\mp(S; s_{\rm eff}) = S^2 - \frac{\sqrt{3}\mp1}{2} s_{\rm eff} S + \frac{s_{\rm eff}^2 - 2 \mp \sqrt{3}}{2}$. (The coefficient $-\frac{1}{24}$ printed in Phase 2g is sixteen times too small because column replacement $M_{\cdot,1} \pm M_{\cdot,2}$ multiplies the determinant by 4.)

2. *At the continuum threshold $S = 1$ ($g = 0$), the two quadratic factors factorise exactly as*

$$P_-(1; s_{\rm eff}) = \frac{1}{2}(s_{\rm eff} + 1)(s_{\rm eff} - \sqrt{3}), \qquad P_+(1; s_{\rm eff}) = \frac{1}{2}(s_{\rm eff} - 1)(s_{\rm eff} - \sqrt{3}). \tag{22}$$

3. *Retaining only admissible decaying half-line roots ($S > 1$ for $s_{\rm eff} \in (-2, 2)$):*

   - *For $s_{\rm eff} \le -1$ ($\eta q_{\rm vac} \ge \frac{1}{2\sqrt{3}}$): **zero** growing modes exist; these are the only linearly stable defect vacua, and the only vacua admissible as external quantum states.*
   - *For $-1 < s_{\rm eff} \le 1$: exactly **one** growing mode exists, given by $S_{-,+}(s_{\rm eff}) > 1$ from $P_- = 0$, with growth rate $g_-(s_{\rm eff}) = \frac{s_{\rm eff}\sqrt{4 - s_{\rm eff}^2} + \sqrt{3}(2 - s_{\rm eff}^2)}{8} > 0$.*
   - *For $1 < s_{\rm eff} < 2$: a second growing mode $S_{+,+}(s_{\rm eff}) > 1$ emerges from $P_+ = 0$ at $s_{\rm eff} = 1$ with growth rate $g_{+,+}(s_{\rm eff}) = \frac{s_{\rm eff}\sqrt{4 - s_{\rm eff}^2} - \sqrt{3}(2 - s_{\rm eff}^2)}{8} > 0$. At $s_{\rm eff} = \sqrt{3}$, both $P_-(1; \sqrt{3}) = 0$ and $P_+(1; \sqrt{3}) = 0$: the root $S_{-,+}(s_{\rm eff})$ of P_- decreases through $S = 1$ ($g = 0$) and leaves the admissible sheet $S > 1$, while the second root $S_{+,-}(s_{\rm eff}) = \frac{(\sqrt{3}+1)s_{\rm eff} - (\sqrt{3}-1)\sqrt{4 - s_{\rm eff}^2}}{4}$ of P_+ simultaneously increases through $S = 1$ and enters the admissible sheet for $\sqrt{3} < s_{\rm eff} < 2$. Thus on $(1, 2)$ the two non-negative admissible growth rates are (verified to 55 digits)*

$$g_1(s_{\rm eff}) = \frac{s_{\rm eff}\sqrt{4 - s_{\rm eff}^2} - \sqrt{3}(2 - s_{\rm eff}^2)}{8} > 0, \quad g_2(s_{\rm eff}) = \frac{\left|s_{\rm eff}\sqrt{4 - s_{\rm eff}^2} + \sqrt{3}(2 - s_{\rm eff}^2)\right|}{8} \ge 0, \tag{23}$$

which both belong to $P_+(S; s_{\rm eff}) = 0$ for $\sqrt{3} < s_{\rm eff} \le 2$ and coalesce at $s_{\rm eff} = 2$ to $g_1(2) = g_2(2) = \frac{\sqrt{3}}{4}$.

3.2 Direct operator derivation of ΔP_D and ΔE_D from $P_{D,2}$ and D_2

Expanding the project-normalised defect Lagrangian (10) and localised momentum (16) to quadratic order in potential fluctuations $(a, b) = (\delta\varphi_L, \delta\varphi_R)$ about a constant vacuum σ_0 gives **(c)** (`audit_charges_action.pdf`, `phase2/scripts/defq_oneloop_renorm.py`):

$$D_2 = a_t F_1 + b_t G_1 - E_{D,2}, \qquad F_1 = \alpha a_x + \beta b_x - \frac{\sigma_0}{2}(a-b), \qquad G_1 = \beta a_x + \alpha b_x - \frac{\sigma_0}{2}(a-b), \tag{24}$$

$$E_{D,2} = -H_2 = -\sigma_0\left[\frac{\beta}{2}(a_x^2+b_x^2) + \alpha a_x b_x\right] - \frac{1-\sigma_0^2}{2}(a-b)(a_x+b_x) + \frac{\beta\sigma_0(1-\sigma_0^2)}{8}(a-b)^2, \tag{25}$$

$$P_{D,2} = -\frac{\alpha}{2}(a_x^2+b_x^2) - \beta a_x b_x - \frac{\alpha}{4}(1-\sigma_0^2)(a-b)^2, \tag{26}$$

where $\alpha = \eta/\sqrt{3}$ and $\beta = 2\alpha$. In `audit_charges_action.pdf` (Section 4), it was noted as an open issue **(d)** that staticity alone does not prove why the expectation value of the conserved momentum including $P_{D,2}$ equals the phase-shift integral $\int_{-\infty}^{\infty} \frac{\mathrm{d}q}{4\pi}\, q\, \theta_D'(q)$. We now supply this derivation **(c)**.

On any constant defect vacuum, the continuum phonon eigenmodes are reflectionless plane waves all the way to $x = 0^{\mp}$ (with zero evanescent tails **(c)**, Claim 2.1 of `oneloop_defect_phase.pdf`). With canonical bulk field normalisation ($[\varphi(x), \pi(y)] = \mathrm{i}\hbar\delta(x-y)$, zero-point weight $\frac{1}{2}\hbar$ per mode), a mode of wavenumber $q \in \mathbb{R}\setminus\{0\}$ and frequency $\omega(q) = |q|\sqrt{1+q^2}$ has

$$\begin{aligned} a_q(x,t) &= \frac{1}{\sqrt{\pi\omega(q)}}\,\mathrm{Re}\big(\mathrm{e}^{\mathrm{i}(qx-\omega(q)t)}\big) \quad (x<0), \\ b_q(x,t) &= \frac{1}{\sqrt{\pi\omega(q)}}\,\mathrm{Re}\big(z_{\mathrm{ph}}(q;q_D,\eta)\,\mathrm{e}^{\mathrm{i}(qx-\omega(q)t)}\big) \quad (x>0), \end{aligned} \tag{27}$$

so that $\langle a^2\rangle_q = \langle b^2\rangle_q = \frac{1}{4\pi\omega(q)}$, $\langle a_x^2\rangle_q = \langle b_x^2\rangle_q = \frac{q^2}{4\pi\omega(q)}$, $\langle a_x b_x\rangle_q = \frac{q^2\,\mathrm{Re}\, z_{\mathrm{ph}}(q)}{4\pi\omega(q)}$, $\langle (a-b)^2\rangle_q = \frac{2(1-\mathrm{Re}\, z_{\mathrm{ph}}(q))}{4\pi\omega(q)}$, and $\langle (a-b)(a_x+b_x)\rangle_q = -\frac{2q\,\mathrm{Im}\, z_{\mathrm{ph}}(q)}{4\pi\omega(q)}$ at $x=0$. Because $|z_{\mathrm{ph}}(q)| = 1$ everywhere on $x>0$, the bulk momentum density $-\varphi_t\varphi_x$ on mode q is spatially constant and identical to its free-vacuum value, so its vacuum-subtracted bulk integral and its $\pm q$ symmetric bulk sum vanish identically. All vacuum-subtracted momentum of the mode pair $(+q,-q)$ is therefore localised in $P_{D,2}$ at $x=0$.

Proposition 3.2 (Exact pointwise identity between $\langle P_{D,2}\rangle_q$ and $\frac{q}{4\pi}\theta_D'(q)$, **(c)**, `phase2/scripts/defq_oneloop_renorm.py`). *For every $q>0$, every $\sigma_0 \in \mathbb{R}$, and $\eta \in \{+1,-1\}$:*

1. *As $q\to\infty$, $z_{\mathrm{ph}}(q;q_D,\eta) \to \mathrm{e}^{-\mathrm{i}\eta\,\mathrm{sgn}(q)\frac{2\pi}{3}}$ has $\mathrm{Re}\, z_{\mathrm{ph}}(q) \to -\frac{1}{2}$. Because $\beta = 2\alpha$, the leading $O(q^0)$ derivative combination $-\frac{\alpha}{2}(a_x^2+b_x^2) - \beta a_x b_x \propto 1 + 2\,\mathrm{Re}\, z_{\mathrm{ph}}(q) \to 0$ cancels identically at $O(1)$, leaving $\langle P_{D,2}\rangle_q = -\frac{\sqrt{3}\eta\sigma_0}{8\pi q} + \frac{\sqrt{3}\eta(\sigma_0^2-2)}{16\pi q^2} + O(q^{-3})$.*

2. *Symmetrising over $+q$ and $-q$ cancels the odd $O(q^{-1})$ tail and gives the **exact pointwise algebraic identity***

$$\boxed{\begin{aligned} \langle P_{D,2}\rangle_q + \langle P_{D,2}\rangle_{-q} &\equiv \frac{q}{4\pi}\Big(\theta_D'(q;s,\eta) - \theta_D'(-q;s,\eta)\Big) \\ &= \frac{\sqrt{3}\,\eta\, q\big(2(\sigma_0^2-2)q^2 + \sigma_0^2 - 1\big)}{\pi\sqrt{1+q^2}\big(16q^4 + 4(\sigma_0^2+2)q^2 + (\sigma_0^2-1)^2\big)}. \end{aligned}} \tag{28}$$

Consequently, the vacuum expectation value of the localised quadratic defect momentum operator is UV finite and equals $\Delta P_D(q_D,\eta)$ identically:

$$\langle P_{D,2}\rangle_{\mathrm{vac}} := \int_{-\infty}^{\infty} \mathrm{d}q\, \langle P_{D,2}\rangle_q = \int_{-\infty}^{\infty} \frac{\mathrm{d}q}{4\pi}\, q\, \theta_D'(q;s,\eta) = \Delta P_D(q_D,\eta). \tag{29}$$

3. *For the quadratic energy, integrating the bulk potential energy* $\frac{1}{2}\int_{x\neq 0}(\varphi_x^2+\varphi_{xx}^2)\,\mathrm{d}x$ *by parts at* $x=0^{\mp}$ *produces the boundary jump* $J_{\mathrm{bulk},2} := \frac{1}{2}[a(a_x-a_{xxx})+a_x a_{xx}]_{0^-} - \frac{1}{2}[b(b_x-b_{xxx})+b_x b_{xx}]_{0^+}$. *By Euler's homogeneous function theorem on* $D_2 = a_t F_1 + b_t G_1 - E_{D,2}$ *and the linearised sewing equations, every harmonic sewing mode satisfies* $\langle E_{D,2}+J_{\mathrm{bulk},2}\rangle_q = \langle a_t F_1 + b_t G_1\rangle_q$, *and the symmetric sum over* $\pm q$ *obeys the* ***exact pointwise identity***

$$\boxed{-\Big(\langle a_t F_1 + b_t G_1\rangle_q + \langle a_t F_1 + b_t G_1\rangle_{-q}\Big) \equiv \frac{\omega(q)}{4\pi}\Big(\theta_D'(q;s,\eta)+\theta_D'(-q;s,\eta)\Big).} \tag{30}$$

3.3 Cutoff Casimir energy, boundary UV divergence, and counterterm analysis

On any linearly stable defect vacuum ($s_{\mathrm{eff}} = \eta\sigma_{\mathrm{vac}} < -1$), there are no bound or growing discrete modes. In the equal-mode-count Gaussian quantisation prescription [2, App. 6] (audited for the bulk soliton in `audit_charges_action.pdf`), integrating the mode-shift sum by parts on $[-\Lambda,0^-]\cup[0^+,\Lambda]$ requires the **asymptotic scattering branch** $\theta_D^{\mathrm{asympt}}(q;s,\eta) := \sum_{s'=1}^{2} c_{D,s'}\Theta(q;\psi_{s'})$, which satisfies $\theta_D^{\mathrm{asympt}}(\pm\infty;s,\eta)=0$ (whereas the zero-endpoint branch $\theta_D^{\mathrm{phys}}(q)$ with $\theta_D^{\mathrm{phys}}(0^{\pm})=0$ approaches $\pm\pi\eta$ at $q\to\pm\infty$ for $s_{\mathrm{eff}}<-1$; its nonzero jump at $q=0^{\pm}$ is multiplied by $\omega(0)=0$ in the energy integration by parts and therefore drops out **(c)**, Section 5, Failure 3).

Proposition 3.3 (Regulated defect Casimir energy and boundary counterterm structure, **(c)**, `phase2/scripts/defq_oneloop_renorm.py`). *Let* $s_{\mathrm{eff}} = \eta\sigma_{\mathrm{vac}} < -1$, $S_D := \sum_{s'=1}^{2} c_{D,s'}\sin\psi_{s'} = -\frac{\eta\sqrt{3}\sigma_{\mathrm{vac}}}{2} = 3\eta q_{\mathrm{vac}} > \frac{\sqrt{3}}{2}$, *and* $T := \sqrt{1+\Lambda^2}$. *Then:*

1. *As* $\Lambda\to\infty$, *the phase-density integral* $E_D^{\mathrm{dens}}(\Lambda)$ *and cutoff endpoint* $B_D(\Lambda)$ *in the asymptotic branch* $\theta_D^{\mathrm{asympt}}(\pm\infty)=0$ *satisfy (verified to* $<10^{-20}$ *at* $\Lambda=10^{22}$*)*

$$E_D^{\mathrm{dens}}(\Lambda) := \int_{-\Lambda}^{\Lambda}\frac{\mathrm{d}q}{4\pi}\,\omega(q)\theta_D'(q) = \frac{S_D}{2\pi}(T-1) + \left(\Delta E_D^{\mathrm{ren}} - \frac{S_D}{2\pi}\right) + O(\Lambda^{-1}), \tag{31}$$

$$B_D(\Lambda) := \frac{\Lambda\sqrt{1+\Lambda^2}}{4\pi}\Big[\theta_D^{\mathrm{asympt}}(\Lambda) - \theta_D^{\mathrm{asympt}}(-\Lambda)\Big] = -\frac{S_D}{2\pi}\,T + O(\Lambda^{-1}), \tag{32}$$

$$E_D^{\mathrm{mode}}(\Lambda) := E_D^{\mathrm{dens}}(\Lambda) - B_D(\Lambda) = \frac{S_D}{\pi}(T-1) + \Delta E_D^{\mathrm{ren}} + O(\Lambda^{-1}), \tag{33}$$

where $\frac{T-1}{2\pi} = \int_{-\Lambda}^{\Lambda}\frac{\mathrm{d}q}{2\pi}\frac{|q|}{2\sqrt{1+q^2}} = \langle\varphi_x^2\rangle_{\mathrm{vac}}(\Lambda)$ *is the exact one-loop bulk tadpole integral and* $\Delta E_D^{\mathrm{ren}}$ *is the finite 2-root dipole formula of Phase 2g. The momentum endpoint* $\frac{\Lambda}{4\pi}[\theta_D^{\mathrm{asympt}}(\Lambda)+\theta_D^{\mathrm{asympt}}(-\Lambda)]$ *vanishes as* $O(\Lambda^{-1})$.

2. ***Need for a boundary counterterm:*** *Because the open-half-line bulk normal-ordering counterterm* $\delta E_{\mathrm{bulk}}^{\mathrm{ct}} = \frac{3(T-1)}{2\pi}\big(\int_{-\infty}^{0}\varphi_{L,x}\,\mathrm{d}x + \int_0^{\infty}\varphi_{R,x}\,\mathrm{d}x\big)$ *vanishes identically on every spatially constant defect vacuum* ($\varphi_{L,x}^{(0)}=\varphi_{R,x}^{(0)}=0$), *cancelling the UV divergence* $-\frac{S_D}{\pi}(T-1) = \frac{\eta\sqrt{3}\sigma_{\mathrm{vac}}}{2\pi}(T-1)$ *of* $E_D^{\mathrm{mode}}(\Lambda)$ *requires an explicit localised boundary counterterm at* $x=0$*:*

$$\delta E_D^{\mathrm{ct}}(\sigma_{\mathrm{vac}};\Lambda) = -\frac{S_D}{\pi}(T-1) + \delta E_{D,\mathrm{fin}}(\sigma_{\mathrm{vac}}) = \frac{\eta\sqrt{3}(T-1)}{2\pi}\big(s+\sqrt{3}\,\eta\,M_D\big) + \delta E_{D,\mathrm{fin}}(\sigma_{\mathrm{vac}}). \tag{34}$$

Notice that the M_D*-dependent part of (34),* $+\frac{3(T-1)}{2\pi}M_D = \frac{3(T-1)}{2\pi}(\varphi_R(0^+)-\varphi_L(0^-))$, *is precisely the boundary jump required to complete the open-half-line bulk integral* $\int_{x\neq0}\varphi_x\,\mathrm{d}x = (\varphi_R(+\infty)-\varphi_L(-\infty)) - M_D$ *into the topological total-derivative counterterm* $\frac{3(T-1)}{2\pi}(\varphi_R(+\infty)-\varphi_L(-\infty))$*!*

3. ***Integrability constraint on finite boundary counterterms:*** *Adding a shift-invariant zero-derivative potential $\delta\tau_{\rm fin}(w_\varphi)$ to $H_{\rm CZ}$ without modifying the derivative sewing terms changes the momentum flux balance on the sewing manifold by $-(u_{{\rm CZ},x}-v_{{\rm CZ},x})\delta\tau'_{\rm fin}(w_{\rm CZ})$, which cannot equal a total time derivative $\frac{\rm d}{{\rm d}t}\delta h_{\rm fin}(w_{\rm CZ}) = (u_{{\rm CZ},t}-v_{{\rm CZ},t})\delta h'_{\rm fin}(w_{\rm CZ})$ for independent boundary jets unless $\delta\tau'_{\rm fin}=\delta h'_{\rm fin}=0$. Thus an off-shell integrable one-loop boundary renormalisation must take the form of a one-loop shift of the integrable CZ defect parameters*

$$(s,\,\rho_0,\,\tau_0,\,h_0)\;\longmapsto\;(s+\hbar\,\delta s_1,\;\rho_0+\hbar\,\delta\rho_{0,1},\;\tau_0+\hbar\,\delta\tau_{0,1},\;h_0+\hbar\,\delta h_{0,1}). \tag{35}$$

*Here $(\delta\tau_{0,1},\delta h_{0,1})$ are vacuum-independent additive constants that cancel in vacuum jumps; keeping $w_\varphi=0$ ($\sigma=s$) as a static vacuum requires $\rho(s)=0$, which ties $\delta\rho_{0,1}$ to δs_1 and leaves **one free finite parameter renormalisation** δs_1 (or two parameters $(\delta s_1,\delta\rho_{0,1})$ if the vacuum jump $w_\varphi^{(0)}=O(\hbar)$ is allowed to shift at one loop).*

4 Quantum absorption and fusion test at $O(\hbar^0)$

We now test what the classical and one-loop boundary charges imply for soliton absorption and KP-triangle defect fusion at $O(\hbar^0)$ (`phase2/scripts/defq_absorption_fusion.py`).

4.1 Obstruction 1: No elementary single-soliton absorption connects two stable vacua

Proposition 4.1 (Complete growing-mode count on KP triples, **(c)**, `phase2/scripts/defq_absorption_fusion.py`). *Parametrise any nondegenerate KP triple $\{q_1,q_2,q_3\}$ ($q_1+q_2+q_3=0$, $q_1q_2+q_2q_3+q_3q_1=-\frac14$) by $q_j=-\frac{1}{\sqrt3}\cos(\vartheta+\frac{2\pi j}{3})$ with $\vartheta\in(0,\pi)\setminus\{\frac\pi3,\frac{2\pi}{3}\}$. For either defect orientation $\eta\in\{+1,-1\}$, the three effective defect parameters $s_{{\rm eff},j}=-2\sqrt3\,\eta\,q_j=2\eta\cos(\vartheta+\frac{2\pi j}{3})$ lie in $(-2,-1)$, $(-1,1)$, and $(1,2)$ respectively. By Proposition 3.1(3):*

1. *The sorted number of strictly positive admissible growing modes ($S>1$) on the three vertices $\{q_1,q_2,q_3\}$ is $(0,1,2)$ for all $\vartheta\not\equiv\frac\pi6\pmod{\frac\pi3}$, and $(0,1,1)$ at the isolated branch-switch angles $\vartheta\equiv\frac\pi6\pmod{\frac\pi3}$ where the $(1,2)$ vertex hits $s_{\rm eff}=\sqrt3$ ($g_2(\sqrt3)=0$ at threshold $S=1$, while $g_1(\sqrt3)=\frac{2\sqrt3-1}{8}\approx0.3080>0$).*
2. *In every nondegenerate KP triple (checked across 594 directed edges), **exactly one vertex $q_{\rm st}$ is linearly stable** ($s_{\rm eff}<-1$), and **both excited vertices are linearly unstable**. Consequently, **no** elementary single-soliton absorption or emission process $D(q_i)+S_i\leftrightarrow D(q_{i+1})$ connects two linearly stable external defect states.*

4.2 Obstruction 2: Non-zero KP-triangle holonomy of canonical collective-coordinate charges

Even if one continues the one-loop defect vacuum charges $(\Delta E_D^{\rm ren}(q_i,\eta),\Delta P_D(q_i,\eta))$ analytically from the stable region $s_{\rm eff}<-1$ to the two unstable vertices of a KP triple, a second obstruction arises from the bulk soliton velocity anomaly (`charge_frame_master.pdf`, `audit_charges_action.pdf`). Recall that for a bulk soliton $S=(\alpha,\epsilon)$, the fixed-profile mode-sum charges $(\Delta E_1^{\rm ren}(\alpha),\Delta P_{\rm mode}(\alpha,\epsilon))$ violate ${\rm d}E=v\,{\rm d}P$ by $\mathcal{A}_v(\alpha)=\frac{{\rm d}\Delta E_1^{\rm ren}}{{\rm d}\alpha}-v\frac{{\rm d}\Delta P_{\rm mode}}{{\rm d}\alpha}\neq0$, whereas the co-moving canonical collective-coordinate derivatives $\Delta P_{\rm can}=\frac{\partial L_1}{\partial v}$ and $\Delta E_{\rm can}=v\Delta P_{\rm can}-L_1$ ($L_1=v\Delta P_{\rm mode}-\Delta E_1^{\rm ren}$) satisfy ${\rm d}\Delta E_{\rm can}=v\,{\rm d}\Delta P_{\rm can}$ **(c)**.

Proposition 4.2 (KP-triangle holonomy no-go theorem for canonical soliton charges, **(c)**, `phase2/scripts/defq_absorption_fusion.py`). *Let $\{q_1,q_2,q_3\}$ be a nondegenerate KP triple with edges $S_i=\{q_i,q_{i+1}\}$ ($i=1,2,3$, $q_4\equiv q_1$) and absorption signs $\rho_i=\eta\,{\rm sgn}(q_i-q_{i+1})$.*

1. *Under minimal tadpole subtraction ($\delta s_1 = 0$), the 2-root dipole defect shifts match the* ***mode-sum*** *soliton charges on every edge, $\Delta(E_D^{\rm ren}, P_D)_{q_i \to q_{i+1}} = \rho_i(\Delta E_1^{\rm ren}(S_i), \Delta P_{\rm mode}(S_i))$, and their directed sum around the closed KP triangle vanishes identically (verified to 8.02×10^{-51}):*

$$\sum_{i=1}^{3} \rho_i \, \Delta E_1^{\rm ren}(S_i) = 0, \qquad \sum_{i=1}^{3} \rho_i \, \Delta P_{\rm mode}(S_i) = 0. \tag{36}$$

2. *By contrast, the co-moving canonical collective-coordinate charges $(\Delta E_{\rm can}, \Delta P_{\rm can})$ have a* ***non-zero directed holonomy around every generic KP triangle****:*

$$\boxed{\mathcal{C}_E(q_1,q_2,q_3) := \sum_{i=1}^{3} \rho_i \, \Delta E_{\rm can}(S_i) \neq 0, \qquad \mathcal{C}_P(q_1,q_2,q_3) := \sum_{i=1}^{3} \rho_i \, \Delta P_{\rm can}(S_i) \neq 0.} \tag{37}$$

   For example, for $\eta = +1$:

KP seed root q_1	$\mathcal{C}_E(q_1,q_2,q_3)$	$\mathcal{C}_P(q_1,q_2,q_3)$
-0.20	$+0.3175373286$	$+0.1655848943$
$+0.10$	$+0.4898832009$	-0.1111527059
$+0.45$	-0.5020709674	$+0.09965754968$

3. ***No-go consequence:*** *Because any single-valued assignment of renormalised defect vacuum charges $(E_D^{\rm quant}(q_i), P_D^{\rm quant}(q_i))$ on the vertices $\{q_1,q_2,q_3\}$—regardless of what finite or divergent boundary counterterms $\delta E_D^{\rm ct}(q_i), \delta P_D^{\rm ct}(q_i)$ are added at the defect—satisfies $\sum_{i=1}^{3} \big(Q_D^{\rm quant}(q_{i+1}) - Q_D^{\rm quant}(q_i)\big) \equiv 0$,* ***no boundary counterterm or defect state-charge assignment can balance*** *$(\Delta E_{\rm can}(S_i), \Delta P_{\rm can}(S_i))$* ***at the unshifted classical absorption rapidities across all three edges of a KP triangle.***

4. *Only if each absorption rapidity shifts at $O(\hbar)$ relative to the defect vertex by the conditional kinematic shift of* `audit_charges_action.pdf`*,*

$$\alpha_{S_i}^{\rm abs} = \alpha_{S_i} + \hbar \, a_{1,i}, \qquad a_{1,i} = \frac{\Delta P_{\rm mode}(S_i) - \Delta P_{\rm can}(S_i)}{P'_{\rm cl}(S_i)}, \tag{38}$$

   do the shifted canonical soliton charges $E_{\rm cl}(S_i) + \hbar(E'_{\rm cl}(S_i)a_{1,i} + \Delta E_{\rm can}(S_i)) = E_{\rm cl}(S_i) + \hbar \Delta E_1^{\rm ren}(S_i)$ and $P_{\rm cl}(S_i) + \hbar(P'_{\rm cl}(S_i)a_{1,i} + \Delta P_{\rm can}(S_i)) = P_{\rm cl}(S_i) + \hbar \Delta P_{\rm mode}(S_i)$ reduce back to the mode-sum charges and close around the triangle (verified to $< 10^{-49}$). Establishing (38) physically would require a one-loop quantum sewing/bound-state pole calculation ***(d)****.*

4.3 Obstruction 3: Absorbed-soliton momentum renormalisation in fixed-momentum fusion

In Phase 2g (`oneloop_defect_phase.pdf`, Claim 6.1), one-loop defect fusion for a test soliton $3 = (\alpha_3, \epsilon_3)$ was written as the sum of the continuum integral identity

$$I_{SD}(3; q_{i+1}, \eta) - I_{SD}(3; q_i, \eta) = \rho_i \, I_{SS}(3, S_i) \tag{39}$$

and the test-soliton momentum-renormalisation shift $-\big(\Delta x_D(3; q_{i+1}, \eta) - \Delta x_D(3; q_i, \eta)\big)\Delta P(3) = -\rho_{{\rm tree},i} \Delta X_3(3, S_i) \Delta P(3)$ ($\rho_{{\rm tree},i} = \epsilon_3 \rho_i$), both of which sum to zero around a closed KP triangle **(c)**.

However, when the two-soliton S-matrix $S(P_3^{\rm phys}, P_{S_i}^{\rm phys})$ is expressed at fixed physical momenta of **both** colliding solitons 3 and S_i, expanding the tree-level phase $\delta_0(3, S_i)$ around $(P_3^{\rm phys}, P_{S_i}^{\rm phys})$

produces two Taylor terms:

$$\begin{aligned}\delta_{12,\mathrm{renorm}}(3,S_i) &= -\frac{\partial\delta_0(3,S_i)}{\partial P_3}\Delta P(3) - \frac{\partial\delta_0(3,S_i)}{\partial P_{S_i}}\Delta P(S_i)\\ &= \Delta X_3(3,S_i)\,\Delta P(3) + \Delta X_{S_i}(3,S_i)\,\Delta P(S_i),\end{aligned} \tag{40}$$

where $\Delta X_{S_i}(3,S_i) = +k_{S_i}^{-1}\ln|A_{12}(3,S_i)|$.

Proposition 4.3 (Non-closure of the absorbed-soliton momentum shift around a KP triangle, **(c)**, `phase2/scripts/defq_absorption_fusion.py`). *Using* $\ln|A_{12}(3,S_i)| = \rho_{\mathrm{tree},i}\big(\ln|z_t(3;q_{i+1},\eta)| - \ln|z_t(3;q_i,\eta)|\big)$ *and summing by parts around any closed KP triangle* $\{q_1,q_2,q_3\}$ *gives the exact identity (verified to* 2.41×10^{-50}*):*

$$\boxed{\sum_{i=1}^{3}\rho_{\mathrm{tree},i}\,\Delta X_{S_i}(3,S_i)\,\Delta P(S_i) = \sum_{i=1}^{3}\left(\frac{\Delta P(S_{i-1})}{k_{S_{i-1}}} - \frac{\Delta P(S_i)}{k_{S_i}}\right)\ln\left|z_t(3;q_i,\eta)\right| \neq 0.} \tag{41}$$

Because $\Delta P(S_i)/k_{S_i}$ *varies across the three edges* S_1,S_2,S_3*, the right-hand side of* (41) *is a non-zero function of the test soliton rapidity* α_3 *with logarithmic singularities at* $p_{3,a/b} = q_i$ *(e.g. at* $q_1 = -0.2, \alpha_3 = 0.35, \epsilon_3 = \eta = +1$*, it equals* -0.7742169719 *for* ΔP_{mode} *and* -1.737010868 *for* ΔP_{can}*). Moreover, because* $\frac{\partial}{\partial q_i}\delta_D^{\mathrm{trans}}(3;q_i,\eta)$ *has simple poles* $\frac{\ell_3-\ell(q_i)}{p_{3,a/b}-q_i}$ *that are absent from* (41)*, no* α_3*-independent one-loop shift* $q_i \mapsto q_i + \hbar\,\delta q_i$ *of the defect vertices can cancel* (41) *for all test solitons* 3*.*

4.4 What survives at $O(\hbar^0)$: precise statement

Combining Propositions 3.1–4.3, we state precisely what holds **(c)** and what does not survive as a physical bootstrap selection rule **(d)**:

1. **Well-posed one-loop transmission on the stable defect ground state (c):** For any defect in its unique linearly stable vacuum q_{st} ($s_{\mathrm{eff}} = \eta\sigma_{\mathrm{st}} < -1$), the linearised spectrum has no growing modes, phonon scattering is reflectionless ($|z_{\mathrm{ph}}(q)| = 1$), the localised momentum operator expectation value equals $\Delta P_D(q_{\mathrm{st}},\eta)$ (Proposition 3.2), the linear Casimir UV divergence is absorbed by the boundary counterterm (34) up to one integrable parameter shift δs_1 (Proposition 3.3), and N-soliton transmission factorises at one loop into pairwise phases.

2. **Surviving algebraic KP-chord identities (c):** At fixed bare KP root parameters $(p_{j,a}, p_{j,b}, q_i)$, the continuum phase integrals obey the exact algebraic chord difference (39) and closed-triangle identity $\sum_{i=1}^{3}\rho_i\, I_{SS}(3,S_i) = 0$, together with test-soliton delay closure $\sum_{i=1}^{3}\rho_{\mathrm{tree},i}\Delta X_3(3,S_i)\Delta P(3) = 0$ and mode-sum charge balance (36). Likewise, a composite two-soliton process $D(q_{\mathrm{st}}) + S_{12} + S_{23} \to D(q_{\mathrm{st}}) + S_{13}$ begins and ends on the unique stable defect vacuum q_{st}.

3. **Why physical KP-triangle fusion does not fix the $O(\hbar^0)$ two-soliton S-matrix (c)/(d):** Promoting the algebraic triangle identities to a physical quantum bootstrap selection rule (e.g. to exclude $\pm\frac{\pi}{2}$ exchange constants or Γ-factors in S_{12}) is obstructed at $O(\hbar^0)$ because: (i) factoring the composite process into elementary steps traverses the two linearly unstable defect vacua q_2, q_3 ($\eta\sigma_{2,3} > -1$), which are not asymptotic quantum states; (ii) canonical soliton charges $(\Delta E_{\mathrm{can}}, \Delta P_{\mathrm{can}})$ have non-zero KP-triangle holonomy (37) unless the absorption locus shifts at $O(\hbar)$ by (38); and (iii) holding the absorbed solitons' physical momenta fixed introduces the non-closing remainder (41).

Claim / Formula	Label	Verification script	Precision / Result
Classical defect charges (E_D, P_D, Π_D, M_D) (12), (16)–(18) and off-shell conservation (Prop. 2.1)	(c)	`phase2/scripts/defq_classical_charges.py`	Exact symbolic SymPy identity
KP-triple vacuum jump factorisation (20) and time-dependent conservation in transmission and absorption (Prop. 2.3)	(c)	`phase2/scripts/defq_classical_charges.py`	Exact symbolic + 50-digit quadrature ($< 10^{-50}$)
Corrected 4×4 sewing determinant prefactor $-\frac{2}{3}$ (21), threshold factorisation (22), and branch switch at $s_{\text{eff}} = \sqrt{3}$ (23) (Prop. 3.1)	(c)	`phase2/scripts/defq_oneloop_renorm.py`	Exact symbolic + 55-digit root check
Pointwise operator identities $\langle P_{D,2}\rangle_{\text{sym}} = \frac{q}{4\pi}\theta'_{D,\text{odd}}$ (28) and $-\langle a_t F_1 + b_t G_1\rangle_{\text{sym}} = \frac{\omega}{4\pi}\theta'_{D,\text{even}}$ (30) (Prop. 3.2)	(c)	`phase2/scripts/defq_oneloop_renorm.py`	Exact symbolic SymPy identity
Cutoff Casimir asymptotics (31)–(33), boundary counterterm (34), and off-shell integrability constraint (Prop. 3.3)	(c)	`phase2/scripts/defq_oneloop_renorm.py`	Exact symbolic + 60-digit check at $\Lambda = 10^{22}$ ($< 10^{-20}$)
Growing-mode count $(0, 1, 2)$ (and $(0, 1, 1)$ at $s_{\text{eff}} = \sqrt{3}$); zero stable-to-stable absorption edges (Prop. 4.1)	(c)	`phase2/scripts/defq_absorption_fusion.py`	Checked across 594 directed KP edges
Non-zero KP-triangle holonomy of $(\Delta E_{\text{can}}, \Delta P_{\text{can}})$ (37) vs zero holonomy of mode-sum and $a_{1,i}$-shifted charges (Prop. 4.2)	(c)	`phase2/scripts/defq_absorption_fusion.py`	50-digit mpmath ($< 10^{-49}$)
Non-closure of absorbed-soliton momentum shift (41) and summation-by-parts identity (Prop. 4.3)	(c)	`phase2/scripts/defq_absorption_fusion.py`	50-digit mpmath ($< 2.5 \times 10^{-50}$)
Quantum derivation of shifted absorption pole (38) and unstable-resonance continuation	(d)	—	Open physical problem

Table 1: Summary of claims, labels, and verification scripts in this write-up.

5 Summary of claims, verification scripts, and failures recorded

Failures recorded. In accordance with `AGENTS.md`, we record all failed checks and errors encountered during this investigation:

1. **Double-subtraction of $-u_{\mathrm{CZ},x}F_{\mathrm{CZ}}-v_{\mathrm{CZ},x}G_{\mathrm{CZ}}$ in P_D and unsimplified $1/\eta$ in SymPy.** In the first version of `phase2/scripts/defq_classical_charges.py`, we defined the total defect momentum as $\frac{1}{9}(-u_{\mathrm{CZ},x}F_{\mathrm{CZ}}-v_{\mathrm{CZ},x}G_{\mathrm{CZ}}+P_{D,\mathrm{CZ}})$ and substituted $\eta^2\mapsto 1$ without simplifying $1/\eta\mapsto\eta$. The off-shell check $\frac{\mathrm{d}}{\mathrm{d}t}(P_{\mathrm{bulk}}+P_D)=0$ failed because CZ's $P_{D,\mathrm{CZ}}$ in [1, (3.49)] already incorporates $-u_{\mathrm{CZ},x}F_{\mathrm{CZ}}-v_{\mathrm{CZ},x}G_{\mathrm{CZ}}+\tilde{P}_D$. Using $P_D=P_{D,\mathrm{CZ}}/9$ and testing $\eta=\pm1$ gave zero off-shell remainder identically.

2. **Sequential dictionary substitution order in SymPy and singular transmission test point.** In Check 2 of `phase2/scripts/defq_classical_charges.py`, a single dict call `.subs({s: s_i, pL: -w_CZ_f/3, eta: eta_val})` left η unreplaced inside `w_CZ_f` when SymPy evaluated η before `pL`; splitting the substitutions fixed the symbolic check. In Check 3(a), our first numerical test point ($s=-1.5,\alpha=0.55,\epsilon=+1,\eta=+1$) had $q_D=\frac{\sqrt{3}}{4}\in(p_b,p_a)$, so $z<0$ (singular soliton on $x>0$) and $\ln z$ was complex, raising a `TypeError` in `mpmath` quadrature; choosing regular transmission points ($s=-2.5,\epsilon=+1$ and $s=-1.5,\epsilon=-1$, where $z>0$) and using tanh / sech jet evaluations to avoid large-argument overflow gave conservation to $<10^{-50}$.

3. **Zero-endpoint vs asymptotic branch in $B_D(\Lambda)$ and sign of $\sigma_0(a-b)/2$ in F_1,G_1.** In the first run of `phase2/scripts/defq_oneloop_renorm.py`, evaluating the cutoff endpoint $B_D(\Lambda)$ with θ_D^{phys} ($\theta_D^{\mathrm{phys}}(0^\pm)=0$) failed because $\theta_D^{\mathrm{phys}}(\pm\infty)=\pm\pi\eta$ for $s<-1$; using the asymptotic branch $\theta_D^{\mathrm{asympt}}(\pm\infty)=0$ (where the $q=0^\pm$ jump is killed by $\omega(0)=0$) and accounting for the $+\frac{S_D}{2\pi}$ rational term in $\Delta E_D^{\mathrm{ren}}$ gave agreement to $<10^{-20}$. In Part 3, writing $+\frac{\sigma_0}{2}(a-b)$ instead of $-\frac{\sigma_0}{2}(a-b)$ in F_1,G_1 broke quadratic momentum conservation $\frac{\mathrm{d}}{\mathrm{d}t}(P_{\mathrm{bulk},2}+P_{D,2})=0$; fixing the sign from $\delta(\frac{\sigma^2}{12\mu})=-\frac{\sigma_0}{2}(a-b)$ and correcting the $-q$ derivative sign in $\theta_D'(-q)$ made all quadratic identities pass identically.

4. **Threshold/branch-switch mode counts in `phase2/scripts/defq_absorption_fusion.py`.** In Check 1 of `phase2/scripts/defq_absorption_fusion.py`, we initially excluded $n=67$ instead of $n=68$ ($2\times102/3=68$) on the grid $\vartheta=\pi n/102$ and expected growing-mode counts $(0,1,2)$ at all nondegenerate angles. The script returned $\{(0,1,2),(0,0,2),(0,1,1)\}$, exposing both the typo $67\to68$ (where $\vartheta=2\pi/3$ is a degenerate threshold triple) and the genuine physical branch-switch point at $n\in\{17,51,85\}$ ($\vartheta\equiv\frac{\pi}{6}\pmod{\frac{\pi}{3}}$, $s_{\mathrm{eff}}=\sqrt{3}$), where the second growing mode touches $g_2(\sqrt{3})=0$ ($S=1$) while $g_1(\sqrt{3})=\frac{2\sqrt{3}-1}{8}>0$ remains strictly unstable, giving mode count $(0,1,1)$ and confirming zero stable-to-stable edges everywhere.

References

[1] E. Corrigan and C. Zambon, *Adding integrable defects to the Boussinesq equation*, J. Phys. A **56** (2023) 405202, arXiv:2306.03772. [arXiv abstract read online for this task; bulk/defect equations and exact solutions rederived and verified symbolically from the action in `corrigan-zambon/defect_check.py`, `corrigan-zambon/verify_defect_solutions.py`, and `phase2/scripts/defq_classical_charges.py`; sections 2–3 and 5.1 previously checked in `audit_charges_action.pdf`.]

[2] L. D. Faddeev and V. E. Korepin, *Quantum theory of solitons*, Phys. Rep. **42** (1978) 1–87, doi:10.1016/0370-1573(78)90058-3. [Relevant equal-mode-count passages in Appendix 6 (eqs. A.6.11–A.6.14) verified in `audit_charges_action.pdf`.]